

Feature	Description
node_outdegree	Total number of distinct outgoing transactions initiated by the account.
node_indegree	Total number of distinct incoming transactions received by the account.
direction_ratio	Ratio of incoming to outgoing transactions (in / (out + ϵ)) a powerful signal of fund-collection behavior in phishing accounts.
max_outgoing_amount	Highest single ETH amount ever sent by the account captures "cash-out" events.
min_outgoing_amount	The smallest non-zero outgoing transaction amount often reveals dust-sweeping patterns.
average_outgoing_amount	Mean value of all outgoing transactions.
max_incoming_amount	The largest single ETH amount ever received is a strong indicator of successful victim exploitation.
min_incoming_amount	Smallest non-zero incoming transaction amount.
average_incoming_amount	Mean value of all incoming transactions.
account_balance	Absolute net flow
account_lifetime	Duration in days between the first and last transaction separates hit-and-run scams from long-running operations.
active_days	The number of distinct calendar days with at least one transaction measures persistence and operational rhythm.
long_term_transfer_frequency	Average daily average number of total transactions in the last 30 days.
short_term_transfer_frequency	The daily average number of total transactions in the last 7 days captures sudden burstiness typical of scams.
long_term_incoming_transfer_frequency	daily average incoming transactions in the last 30 days.
short_term_incoming_transfer_frequency	daily average incoming transactions in the last 7 days is one of the strongest phishing discriminators.
long_term_outgoing_transfer_frequency	daily average outgoing transactions in the last 30 days.
short_term_outgoing_transfer_frequency	daily average outgoing transactions in the last 7 days.
mean_hour_sent	Average clock hour (0–23) of outgoing transactions reveals bot-like or human patterns.
std_hour_sent	Standard deviation of sending hours low values indicate automated scripted behavior.
mean_hour_received	Average clock hour of incoming transactions.
std_hour_received	Standard deviation of receiving hours phishing accounts often show tight clustering around victim active times.
avg_time_between_tx	Mean interval (seconds) between consecutive transactions of the account.
min_time_between_tx	Shortest intervals between any two transactions near-zero values suggest high-frequency bots or ponzi schemes.
max_time_between_tx	Longest inactivity gap useful for identifying dormant-then-burst accounts.
weekend_tx_ratio_sent	Fraction of outgoing transactions occurring on weekends (Sat–Sun).
weekend_tx_ratio_received	Fraction of incoming transactions occurring on weekends many victims transfer on weekends.
katz_centrality	Global influence score considering all possible paths with exponential decay ($\alpha = 0.01$) robust to super-nodes.
degree_centrality	Normalized total degree / (n–1) raw connectivity strength.
closeness_centrality	Reciprocal of average shortest-path distance to all other nodes measures information/flow reachability.
clustering_coefficient	Fraction of triangles around the node high values typical of mixer/contract clusters, low in phishing sinks.
eigenvector_centrality	Influence score derived from the principal eigenvector prioritizes connection to other influential nodes.
indegree_centrality	Normalized incoming degree direct measure of "money attraction" power.
outdegree_centrality	Normalized outgoing degree indicates spending/distribution activity.